

Xiaoxiao YU

Agent Engineer | Full-Stack Engineer

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Full-stack agent engineer who ships production AI products end to end, from React/TypeScript frontends and Python/Node.js backends to the agent runtime underneath. At Viciking (full-time since July 2026), built the agent products that **doubled video-production efficiency**. Before: a browser-agent startup in Ant Group's MaShang AI Accelerator, zkWasm apps with thousands of users, and React Native apps.

Experience

Viciking (Beijing Weixi Network Technology)

Jul 2026 – Present

Agent Engineer, full-time | sole engineer, frontend to agent runtime, on four internal AI production products

- Versus manual editing, **production efficiency doubled**, with **~90% of automated outputs needing no manual re-editing**; dozens of internal users across multiple business lines.
- WeiChuang Drama Studio** (React 19, FastAPI, Temporal, PostgreSQL): a 1-minute episode goes **from script to finished video in 15–30 min**, with episodes in parallel; **~90% need no manual re-editing**, the rest only light edits. Key design: a resumable coordinator (frozen inputs, idempotent compare-and-set writes, partial-success reruns), model fallback and voice repair, and per-call model traces to diagnose slow or stuck calls. ~5,000 tests.
- Weixi Studio** (TypeScript, React, Remotion, PostgreSQL, MCP): turned OpenChatCut, a local-first single-user agentic video editor, into a **multi-tenant SaaS used across business lines**. Key design: a durable agent runtime (Postgres leases with fencing tokens, checkpointed turns, per-call idempotency keys; paid or irreversible tools escalate to a human) and a revision-safe MCP server so external agents edit the same timeline.
- Screenplay Agent Studio** (FastAPI, Pi agent SDK, OpenTelemetry): my eval harness showed optional specialist agents cost **3.7× tokens for no quality gain**, so each of the 6 writing stages stays single-lead (13 role agents in all). Cache-stable prefixes, file-backed state that survives compaction, human-approved edits, and OpenTelemetry alerts on failure ratio and stalled decisions.
- Creative Engine** (TanStack Start, pgvector): writer agent with 9 least-privilege tools over provenance-labeled context; 14 ablation/LLM-judge experiments decided which prompt mechanisms stayed.

Delphinus Lab

Nov 2024 – Jul 2026

Developer Relations Engineer → Ecosystem Director (Jun 2025) | Remote, concurrent with MPhil

- Rebuilt the zkWasm browser SDK from scratch (unified wallet context, L1/L2 signing and deposits), wrote the dApp-starter CLI, and maintained the TypeScript mini-rollup services.
- Solo-built staking, launchpad, prediction-market and social-binder apps, plus the zkWasm Hub site and developer docs; ran DevOps for 8+ ecosystem apps, which reached **thousands of users** and **multi-million-USD TVL**.
- Led the team that built a hosted Kubernetes deployment platform (GitHub integration, Helm jobs, async status/log tracking, CI/CD), hosting **10+ projects**.

Earlier roles (2021–2024, alongside undergraduate studies)

- Rosen** — Founding Full-Stack Engineer (2022–23): React / React Native apps for iOS, Android and Web; backed by PayPal and Circle. **2PM.Network** — Co-Founder & CEO (2024). **HashPunk** — Product Manager (2021–22), raised US\$1M. **1CIRCLE** — Full-Stack Engineer (2021–22). **Conflux** — Intern (2021).

Education

The Hong Kong University of Science and Technology (Guangzhou)

Sep 2024 – Oct 2026

Red Bird MPhil in Data Science and Analytics | graduated Oct 2026

The University of Edinburgh

Sep 2020 – Jun 2024

MA in Cognitive Science, First Class Honours | Exchange: Fudan University, Computer Science (2022–23)

Selected Projects

Resona — Browser Agents for Product Demos

May – Jun 2026

Own startup, sole engineer | Ant Group MaShang AI Accelerator; trial customers | Stagehand, Playwright, AgentBay, BullMQ

- URL in, demo video out: a breadth scan, then parallel per-section depth agents explore the product in AgentBay cloud sandboxes, with a coverage ledger and retry rounds, a risk gate before data-changing actions, hard payment blocks and memory compaction.
- Human login handoff over CDP: the user signs in on the same browser profile, then the agent restores the session and continues. Explore, record and render run as BullMQ jobs; ~500 tests.
- GSAP and Remotion render agents turn the recordings into the finished demo, behind one OpenAI / Anthropic / Gemini gateway with per-purpose model choice.

Viciking Production Tools

Aug – Oct 2026

In production | C# / .NET, SQLite; FastAPI, React, RF-DETR, TensorRT, FFmpeg

- Unattended ad-platform uploader:** batch-uploads drama assets for review. Every irreversible write persists its

intent and takes a one-time authorization first; lost receipts trigger read-only reconciliation, never a blind resubmit; failing dramas are quarantined while the queue continues.

- **GPU video redaction:** detects and masks 13 object classes in production videos. Built the Label Studio annotation loop, RF-DETR fine-tuning with TensorRT export, a lease-based job API, and a timeline review UI.

Novel-to-Script-Team — Multi-Agent Adaptation Harness 175 ★ 33 forks

Open source, sole author | Agent Skills, Python, ChromaDB | for Claude Code / Codex

- Designed 17 role agents and 25 skills that take a novel through ingest, analysis, episode planning, writing, review and storyboard / video prompts. A review director gates every episode, and failures loop back for rewrite and re-review.
- Kept context bounded: the next phase is derived from which files exist, and per-project agent state lets work resume within an episode and reset between episodes to avoid context overflow. Hybrid ChromaDB + jieba TF-IDF retrieval grounds writing in reference scripts.

ShotLoom — AI Film Shot-Breakdown Workspace

Open source, sole author | FastAPI, Redis/RQ, React, TypeScript, Qwen vision + ASR

- Built a shot-by-shot film breakdown: scene detection and clip cutting, then Qwen vision analysis of shot size, camera movement, lighting and narrative function, aligned with ASR transcripts and a whole-film continuity report.
- Routed long videos to whole-video or overlapping chunked passes with bounded model concurrency and retries; RQ workers with refresh-safe SSE progress; Excel/PDF export; 107 end-to-end tests.

RingChat — Agent-Native Messaging Jan – Mar 2026

Team lead, client engineer | Kotlin / Jetpack Compose, Web, iOS | OpenClaw (self-hosted agent gateway)

- Led a **6-person team** building a cross-platform IM in which every user gets a personal AI agent, and worked mainly on the clients (now continued by another team as Gold House). On Android, built the chat UI, optimistic media upload, contact and group search, unread badges and multi-device sync.

Skills

Agents	Context engineering (cache-stable prefixes, compaction, file-backed state), tool/MCP design, multi-agent orchestration, browser agents (Stagehand, Playwright, CDP, AgentBay), human-in-the-loop, evals and LLM-as-judge
Frontend & mobile	TypeScript, React 19, TanStack (Start / Query), Next.js, Vite, Remotion; React Native, Kotlin / Jetpack Compose
Backend	Python (FastAPI), Node.js; PostgreSQL / pgvector (SKIP LOCKED queues), Redis, Temporal, BullMQ, SSE, WebSocket
Infra & models	Docker, Kubernetes, CI/CD, OpenTelemetry; OpenAI, Anthropic, Gemini, Qwen, Doubao/Seedance via Vercel AI SDK and gateways
Other	Solidity, Rust, C++ (ESP32), PyTorch, scikit-learn

Publication

Xiaoxiao YU, L. Du, C. Zeng, L. Deng, H. Luo, X. Gao, J. Tang. *SocialAttributionQA: A Benchmark for Attribution-Evidence-Aware Social Web Question Answering*. **WISE 2026** (CCF B), accepted; Springer LNCS. First author: built a 1,200-question benchmark over 725k Mastodon posts and the ATLAS graph-plus-retrieval solver (**95.3%** accuracy), with ablations and error analysis.

Selected Hackathons & Honours

33 hackathon and competition awards (12 first place or champion); AI and full-stack highlights:

Ava Box — Voice-First Trading Agent **AVE CLAW Hackathon 2026 — Champion**

Sole developer | ESP32-S3, C++, Python, LVGL

- Built a handheld voice agent that routes speech to trading tools based on the current screen; every order needs a button confirmation, and the LVGL UI is shared between firmware and a desktop simulator. Extracted into the open DeviceKit framework (pluggable chain/ASR/LLM/TTS adapters, OTA).

OutcomeX — AI Delivery & Settlement **HashKey Chain Horizon 2026 — 2nd, DeFi Track**

Sole developer | FastAPI, AgentSkillOS (skill-DAG agent runtime), Solidity

- Built a marketplace where paid AI jobs run in an AgentSkillOS subprocess that streams DAG status, logs and artifacts back to the app, with escrow, refunds and revenue claims in 7 separate Foundry contracts.

2025	Zama Bounty, Universal FHEVM SDK — Winner	2024	BeWater AI Crypto — 1st, Security & Privacy
2025	ICBC Cup FinTech, Guangdong — First Prize	2024	ETH Beijing — 1st, Scroll Track
2024	ETH Shanghai — Overall Champion	2021	PaddlePaddle Anti-Fraud ML — 1st (solo)

Founded LambdaSoc (Edinburgh) and Lambda Web3 (**1,000+ members**); **7,000+** Xiaohongshu followers for engineering content. Languages: Chinese (native), English (professional).